TABLE OF CONTENTS

SECTION	SUBJECT	PAGE
1	INTRODUCTION	1-1
2	INSPECTIONS	2-1
3	WEIGHT AND BALANCE	3-1
4	GROUND HANDLING AND SERVICING	4-1
5	AIRFRAME	5-1
6	ELECTRICAL	6-1
7	MAIN ROTOR	7-1
8	MAIN ROTOR CONTROLS	8-1
9	MAST ASSEMBLY	9-1
10	TRANSMISSION	10-1
11	MAIN DRIVE SHAFT	11-1
12	ANTI-TORQUE DRIVE	12-1
13	TAIL ROTOR	13-1
14	POWERPLANT	14-1
15	FUEL SYSTEM	15-1
16	HYDRAULIC SYSTEM	16-1
17	FLIGHT CONTROLS	17-1
18	TOOL LIST	18-1

SHOP ACTIVITIES

In order to provide an atmosphere conducive to actual working conditions,participants in shop activities will be required to maintain a clean working area, and demonstrate safe working characteristics. Eye and foot protection will be required during some shop operations in accordance with OSHA regulations. BHTI will provide the required protective equipment. Performance is carefully monitored and evaluated, allowing immediate feedback of student progress.

INTERACTIVE CD-ROM

The training academy offers a limited selection of training CD-ROM disks for licensed use. Call (817) 280-3047 for a listing of available TRAINING DISKS programs and pricing.

SECURITY REGULATIONS

Students will be issued a Customer Training Academy Identification Badge on the first day of class. This badge must be worn to gain entry to the Bell Cafeteria located adjacent to the Training Academy in the Logistics Center. The Training Academy badge will not permit entry to any other plant or facility at Bell.

FEDERAL SECURITY REGULATIONS DO NOT PERMIT PERSONAL CAMERAS, TAPE RECORDERS, OR FIREARMS AT BELL FACILITIES.

CLASS HOURS

8:00 a.m. - 4:00 p.m. daily (including 1st. & last day of class)

LUNCH BREAK

Classes break for lunch at 12:00 a.m. The Bell Lunch Room offers a catered meal. Vending machines with coffee, soft drinks, candy & snacks are located in the break area. A microwave oven is available in this area as well.

PHONE CALLS

Courtesy phones located in the front lobby, break areas and the pilot lounge area may be used for local and credit card calls. Two pay phones are located in the student's break areas for long distance calls. EMERGENCY calls may be received in the school Administration Office: 817/280-3047.

DRESS CODE

Proper attire is required at all times. Casual clothing such as jeans are acceptable. NO SANDALS / OPEN TOED SHOES, TANK TOPS ARE ALLOWED.

VISITORS

Please be advised that as we are a defense contractor installation, visits by family members (especially children) are restricted by BHTI *Security Department. Prior authorization must be obtained for visits by family members.

SMOKING POLICY

Smoking is prohibited inside all BHTI facilities. Smoking is permitted in designated outside areas <u>only</u>.

MEDICAL CARE

Students must make their own financial arrangements where medical care is concerned.

TRAINING MATERIALS

All supplies, training materials and tools required for training are furnished by the Customer Training Academy. Training materials are provided to <u>enrolled students only</u>.

AIRLINE RESERVATIONS

BHTI personnel are not responsible for making, changing or confirming student airline reservations, nor liable for any costs incurred for airline tickets. DFW International Airport is nearest the training school.

American Airline, 1-800-433-7300
British Airways, 1-800-247-9297
Continental Air Lines, 1-800-525-0280
Delta Air Lines, 1-800-221-1212
KLM Royal Dutch Airline, 1-800-374-7747
Lufthansa Air Lines, 1-800-645-3880
Malaysian Airline, 1-800-552-9264
Mexicana De Aviation S.A., 1-800-531-7921
United Air Lines, U.S. 1-800-241-6522, Intl. 1-800-538-2929
Thai Airways Intl. , 1-800-426-5204
Saudi Arabian Airlines, 1-800-472-8342
Swiss Air, 1-800-221-4750
Singapore Airlines, 1-800-742-3333

AUTO RENTAL

BHTI is not responsible or liable for the reservation of rental cars and/or costs incurred by customers. Most rental agencies require a valid U.S. or International Drivers License and a major credit card.

Advantage Rent-A-Car, 817-284-1100
Avis Car Rental, 1-800-331-1212
Economy Car Rental, 817-640-7368
Snappy Car Rental, 817-485-8808
Alamo Car Rental, 1-800-327-9633
Budget Car Rental, (Metro) 329-2277
Hertz Car Rental, 1-800-654-3131

PROMOTIONAL ITEMS

A limited number of BHTI promotional items (caps, tee-shirts, lapel pins, etc.) are available for sale on Tuesday 9:00 - 4:00 and Thursday 9:00 - 4:00. Items are displayed in the front lobby. Purchases may be made with cash, traveler checks, or personal check if within the U.S.

HOUSING

THE BELL TRAINING ACADEMY CAN NOT GUARANTEE ROOM RESERVATIONS, AND IS NOT LIABLE IN ANY WAY FOR HOTEL ACCOMMODATIONS, SERVICES, AND/OR COSTS INCURRED BY STUDENTS WHILE ATTENDING THE BELL CUSTOMER TRAINING ACADEMY.

TUITION PAYMENT

If tuition has not been prepaid, tuition must be paid on the first day of class in Administration Office. Tuition may be paid in cash, travelers checks, cashier's check, money order, Visa or Master Card.

USA AND CANADA:	1-800-368-BELL (1-800-368-2355)
MEXICO:	95-800-368-2355
ELSEWHERE:	1-817-280-3047/8222

STUDENT NAME: _____

LOCAL ADDRESS: _____

LOCAL PHONE NUMBER: _____

PERMANENT ADDRESS: _____

_____

_____

INSTRUCTORS: _____

_____

412	**33001- 33107**
412SP	**33108- 33213**
412HP	**33214- 33999 & 36020-36086**
412EP	**36087 & SUBSEQUENT**
412CF	**46400- 46499**

PRODUCT SUPPORT ENGINEERING MEDIUM HELICOPTERS

USA: 1-800-363-8028
CANADA: 1-800-363-8224
WORLDWIDE: 450-437-6201
FAX: 450-433-0272

BELL *412 SERIES* Field Maintenance Course Syllabus

1.Course Description: This is an 120 hour course that enables a certificated airframe mechanic to meet the Federal Aviation Administration requirements of FAR Part 65.81, General Privileges and Limitations on the BELL *412 SERIES* helicopter.

2:Course Concept: Acceptable airworthiness methods, techniques and practices as established by governing aviation authorities and Bell Helicopter shall be used as a satisfactory performance level. The course presentation involves classroom and hands-on experience in the shop. The student is required to have the basic skill level of one or more of the following:

(a) Certificated airframe mechanic with one year practical experience as a rated aircraft mechanic.
(b) One year active experience as an active mechanic on helicopters commercial or military.
(c) Three years general experience as an aircraft mechanic commercial or military.

3.Course Objectives: At successful completion, the student will be able to troubleshoot, inspect, perform or supervise the maintenance and/or alteration of the Bell *412 SERIES* up to but not including major repair or overhaul of the helicopter and it's components.

4.Course Outline: The following Table of Contents list the subject number of topics to be discussed, the Class Schedule indicate the number of hours devoted to each subject.

5. Evaluation: Three examinations will be given during the course. The tests will each be a fifty question, multiple choice examination. For successful completion, the student will be required to have an average of 70% or better for the three examinations. In addition to the written examinations, the instructor will administer subject review quizzes at the end of each block of instruction which will not be graded. The student will be graded on his shopwork with either a satisfactory or unsatisfactory grade being given. If the student meets the above requirements, he will be awarded a Certificate of Achievement. If he fails to meet these requirements or is unable to attend all of the course, he will be awarded a Certificate of Attendance.

6. **Recommended References:** The following manuals will be referred to during this course:

Manuals	**412 Maintenance Manuals**	**412 Component Repair & Overhaul**
	412 Structure Repair Manual	**412 Flight Manual**
	BHTI Special Tools Catalog	**BHTI Standard Practices Manual**

BELL COMMERCIAL PUBLICATIONS
GENERAL INFORMATION

Bell Helicopter Textron, Inc. (BHTI) commercial model manuals have certain requirements that determine content and format. Some Flight Manual data is required by regulations of the Federal Aviation Administration, and the Maintenance Manual format is determined by the ATA - 100 Specification of the American Transport Association of America. Some deviations to the specification are made to better adapt the publications to helicopter maintenance.

The instructions set forth in the Flight and Maintenance publications, as supplemented or modified by Alert Service Bulletins and other directions issued by BHTI and Airworthiness Directives issued by the Federal Aviation Administration, must be strictly followed.

Before attempting any procedures not described in the BHTI publications or allied publications, contact Bell Helicopter Textron, Inc. Product Support Department.

HELICOPTER RECORDS

The helicopter records consist of the logbook to enter discrepancies, keep a running total on aircraft operation, provide a record of compliance with required maintenance action, weight and balance data, and historical service records (Overhaul and Retirement Life Components). The engine logbook and engine operating manual are part of the helicopter records and are provided by the engine manufacturer.

FLIGHT MANUAL (FM)

The Flight Manual contains limitations, normal and emergency procedures, and performance data. The BHTI Flight Manual has two parts. Part one is data that is FAA approved and part two is manufacturer's data. The Flight Manual must be carried in the helicopter at all times.

STANDARD PRACTICES MANUAL (SPM)

This manual contains maintenance information and procedures that are common to all Bell Helicopters. Typical information is standard torque charts and application procedures, nondestructive inspection materials and techniques, application of protective finishes, cleaning, thermal fitting procedures, bearing and sleeve replacement, and consumable materials.

STRUCTURAL REPAIR MANUAL (SRM)

The structural repair manual comes in two (2) versions, 206 and MED. Both of these manuals are intended to facilitate field repair of the Bell Helicopter Textron Light and Medium Helicopter Series airframe structure where damage is judged to fall within limits described in this manual. Typical information in both manuals are guidelines for repairs, insert data, rivet replacement information, repair materials and consumable materials.

COMPONENT REPAIR AND OVERHAUL MANUAL

The Component Repair and Overhaul Manual provides data for repair and overhaul of components and assemblies removed from the helicopter. The removal and installation procedures for components and assemblies contained in the Component Repair and Overhaul Manual are provided in the Maintenance Manual.

COMPONENT REPAIR AND OVERHAUL MANUAL VENDOR DATA

This manual contains data for repair and overhaul of vendor supplied components on specific models of Bell Helicopters. The vendor data has been prepared by the manufacturers of those components. It is supplied to owners/operators as a convenient source of component maintenance data. BHTI assumes no responsibility for its accuracy or currency.

MAINTENANCE MANUAL

The Maintenance Manual includes the necessary data to provide helicopter operators
with field maintenance procedures, airworthiness limitations, inspections and general information on the aircraft and systems.

MAINTENANCE MANUALS ON CD.

Avantext Inc.
340 Morgantown Rd.
Reading, PA 19611 USA
Phone: 1-800-998-8857
www.avantext.com

COMPUTERIZED MAINTENANCE RECORD SYSTEM EXAMPLES

Computerized Aircraft Log Manager (CALM)
Attn: Mark Stevens
1955 Raymond Drive, Suite 110
Northbrook, IL 60062
Phone: 1-800-545-2256

Silicon Wings
Attn: Jacqueline Kopec
#212, 4603 Varsity Drive N.W.
Alberta, Canada T3A 2V7
Phone: 1-403-286-8804

MALFUNCTION REPORT (MR)

The MR is a pre-printed form to be submitted to BHTI for all warranty consideration. An MR is also required for all parts returned to BHTI for repair, analysis, or any other reason. The MR can also be used to report any malfunction with a BHTI helicopter. Detailed explanation for use of the MR is provided on the back of the MR form.

OPERATIONS SAFETY NOTICE (OSN)

This notice is approved by the BHTI Safety Board and notifies operators of safety/potential safety conditions that have come to the attention of BHTI. Normally, technical compliance information will not be published in an OSN. Alert Service Bulletins and/or Technical Bulletins are used for this purpose.

INFORMATION LETTER

This publication is used to provide information not covered in other publications and is divided into two major areas:

GENERAL: Not helicopter specific.

SPECIFIC: Helicopter model specific.

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SPECIAL TOOLS ILLUSTRATED PARTS CATALOG (SPECTOOL-IPC)

This publication lists the peculiar and special tools that are applicable to Bell Helicopter Textron's (BHTI) commercial model helicopters. Detail parts of tool assemblies that are procurable from BHTI are included.

ILLUSTRATED PARTS BREAKDOWN (IPB)

The IPB list all parts considered necessary for maintenance repair or overhaul of the helicopter. Commonly used hardware and material, e.g., rivets, cotter pins, grease, adhesives, etc., normally available locally, have been excluded. Explanation on the use of the IPB is contained in the General Information Section of the IPB. The use of parts listed in the IPB may be restricted or prohibited by BHTI Alert Service Bulletins. Accordingly, this manual must be used in conjunction with all applicable BHTI publications. On specific models the IPB is an integral part of the maintenance manual.

ALERT SERVICE BULLETIN (ASB)

This is a document approved by the BHTI Safety Board that notifies operators of matters that affect safety of flight. Compliance with all Alert Service Bulletins is considered mandatory by BHTI.
A list of all ASB's issued is located in Chapter 1, Volume I, of the Maintenance Manual. This list is current as of the date of the manual or its latest revision

TECHNICAL BULLETIN (TB)

This is a document used to provide operators with information of a technical nature that does not affect safety of flight. Examples of subjects covered in a TB are modifications to improve economy or facilitate maintenance, substitution of interchangeable parts, increase in time between inspection and/or overhaul and increase of service and/or retirement time of components.

SERVICE INSTRUCTIONS (SI)

Service Instructions are publications that cover BHTI kits installed on the helicopter. Kits are components or systems that are not part of the basic helicopter. The SI will contain the necessary information to operate and maintain the kit as installed on the helicopter.

UNAPPROVED PARTS
ARE THEY WORTH THE AIRWORTHINESS RISK ??

FOR ASSISTANCE IN DETERMINING THE AUTHENTICITY OF

CALL OUR HOT LINE NUMBER:

1 - 800 - BELL250 *

Or write:

PRODUCT ASSURANCE
BELL HELICOPTER TEXTRON
P.O. BOX 482
FORT WORTH, TEXAS 76101

***If 1-800 - number is not available, call collect 817-280-3118**

BELL'S DEFINITION OF BELL APPROVED PARTS

A Bell-Approved Part is a part that has been purchased from Bell Helicopter Textron, its authorized independent representatives or Bell Customer Service Facilities (CSF).

A Bell-Approved Part will have:

1. **A Bell or vendor part number and an inspection stamp on the part.**

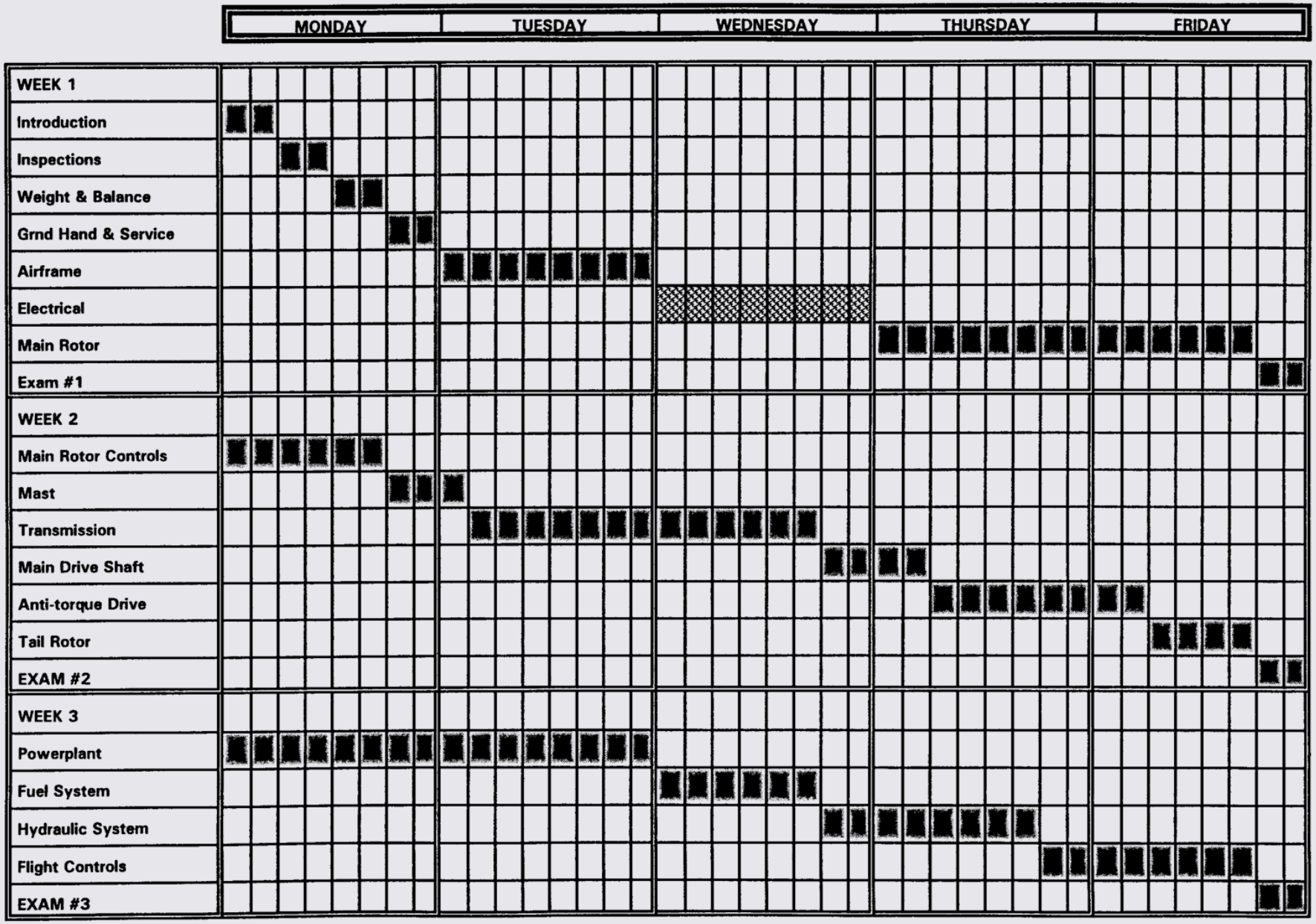
2. **When applicable, metal identification tags with the part number and serial number or part number and applicable serial number permanently applied; and a Bell historical service record if a component has a finite life or scheduled overhaul life.**

3. **The invoice contains a statement signed by a representative of Bell Helicopter Textron, the CSF or independent representative which unequivocally states that the specific part you are purchasing was made under the surveillance, and authority of Bell.**

Other parts that may be installed:

New part manufactured by Bell Helicopter Textron, Inc. and surplused by the U.S. Government may be considered Bell-Approved when they are purchased directly from the U.S. Government in the original Bell Helicopter Textron packing, unopened with seals intact, with United States Government contract identification and proof of manufacture by Bell Helicopter Textron.

An engine part must be purchased from the engine manufacturer or the manufacturer's approved distributor.


MONDAY
TUESDAY
WEDNESDAY
THURSDAY
FRIDAY
WEEK 1
Introduction
Inspections
Weight & Balance
Grnd Hand & Service
Airframe
Electrical
Main Rotor
Exam #1
WEEK 2
Main Rotor Controls
Mast
Transmission
Main Drive Shaft
Anti-torque Drive
Tail Rotor
EXAM #2
WEEK 3
Powerplant
Fuel System
Hydraulic System
Flight Controls
EXAM #3

PREFACE